

# LOUIE CAI

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## EXPERIENCE

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### Software Engineer

Feb 2025 – Present

#### Stripe

- Build software infrastructure, including creation, modification, and testing of source code in Ruby, JavaScript, and JVM languages.
- Monitor production web services for abnormal behavior and respond to incidents.

### Software Development Engineer Intern

Jun – Sept 2024

#### Amazon

- Built a full-stack portal (React + AWS Lambda, Step Functions, DynamoDB) automating internal experiment creation and tracking — cut deployment from 2–3 days to minutes.
- Managed infrastructure with AWS CDK and CloudFormation for seamless deployment and monitoring.

### Software Development Engineer Intern

Jun – Sept 2023

#### Amazon

- Built a React + TypeScript internal tool that cut backend database onboarding from 2–3 days to several hours.
- Collaborated in an Agile team to deliver responsive, user-friendly interfaces.

### CSE Tutor

Sept 2023 – Dec 2024

#### UC San Diego

- Held office hours for CSE 8A and CSE 140L; designed assignments and exams, and automated grade calculation with Python.

## EDUCATION

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### University of California, San Diego, Data Science B.S.

Graduation: Dec 2024

- **GPA:** 3.7
- **Minor:** Computer Engineering
- **Relevant Coursework:** Machine Learning, Deep Learning, NLP, Data Analysis, Computer Organization and Systems Programming, Data Structures & Object-Oriented Design, Differential Equations, Linear Algebra, Probability and Statistics, Discrete Mathematics, Digital Systems Design

## PROJECTS

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### Minecraft Netherite Horse Armor Mod ([CurseForge](#)) ([GitHub](#))

Jun 2020

- Built a Java/Minecraft Forge mod adding Netherite Horse Armor; organically grew to over **2.8 million downloads** on CurseForge.

## ACTIVITIES

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### IEEE F1TENTH IV2023 Robotics Championship - Anchorage ([Website](#))

Jun 2023

- Built two autonomous 1/10-scale RC cars with LiDAR and cameras; led navigation algorithm development in ROS2 and DonkeyCar.
- Represented UC San Diego at the international championship — placed **4th**.

## SKILLS

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- **Languages:** Python, TypeScript, JavaScript, Ruby, Java, C, C++, C#, R, SQL, Bash, SystemVerilog
- **Web:** React, Vite, FastAPI, Spring Boot, HTML, CSS, React Hook Form, React Router DOM
- **ML / Data:** PyTorch, pandas, NumPy, scikit-learn, matplotlib, Selenium, Jupyter
- **Cloud / DevOps:** AWS Lambda, AWS Step Functions, AWS DynamoDB, AWS CDK, AWS CloudFormation, Docker
- **Tools:** Git, Linux, Vim, Unity, Blender, Gradle, JavaFX, Valgrind, Verilator